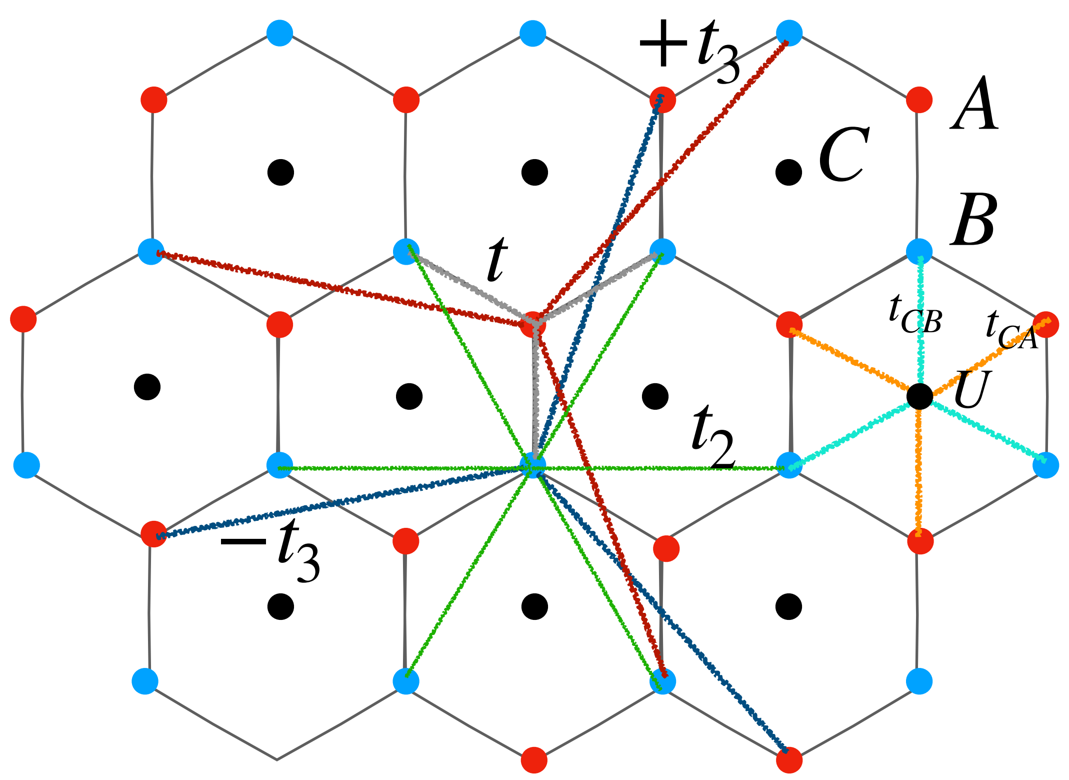
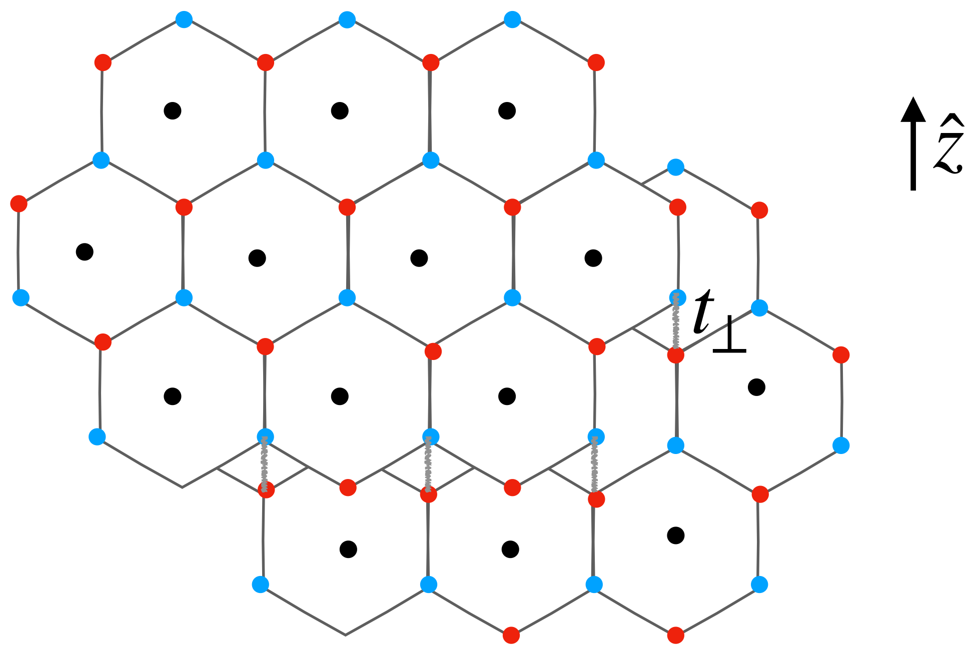
The three band ferroaxial model





$$a_1 = a/2 * [3,\sqrt{3},0]$$
$$a2 = a/2 * [3,-\sqrt{3},0]$$
$$a3 = a * [0,0,1]$$
$$a = 1\text{Å}$$

$$\begin{array}{cccc} & A & B & C \\ H_{3bands} = & \multicolumn{3}{c}{\begin{pmatrix} \multicolumn{2}{c}{H_{2bands}} & h.c. \\ & & h.c. \\ t_{CA}f_{CA}(k) & t_{CB}f_{CB}(k) & -\mu + U \end{pmatrix}} \end{array}$$

$f_{CA}(k) = f(k)$ 1st neighbour hoppings C-A sites

$f_{CB}(k) = f(k)^*$ 1st neighbour hoppings C-B sites

U Onsite energy for C orbitals

$$H_{2bands} = \begin{pmatrix} -\mu + \Delta + t_2 f_2(k) & t\,f(k) + t_3\,g(k) + t_\perp m(k) \\ h.c. & -\mu - \Delta + t_2 f_2(k) \end{pmatrix}$$

Same as before

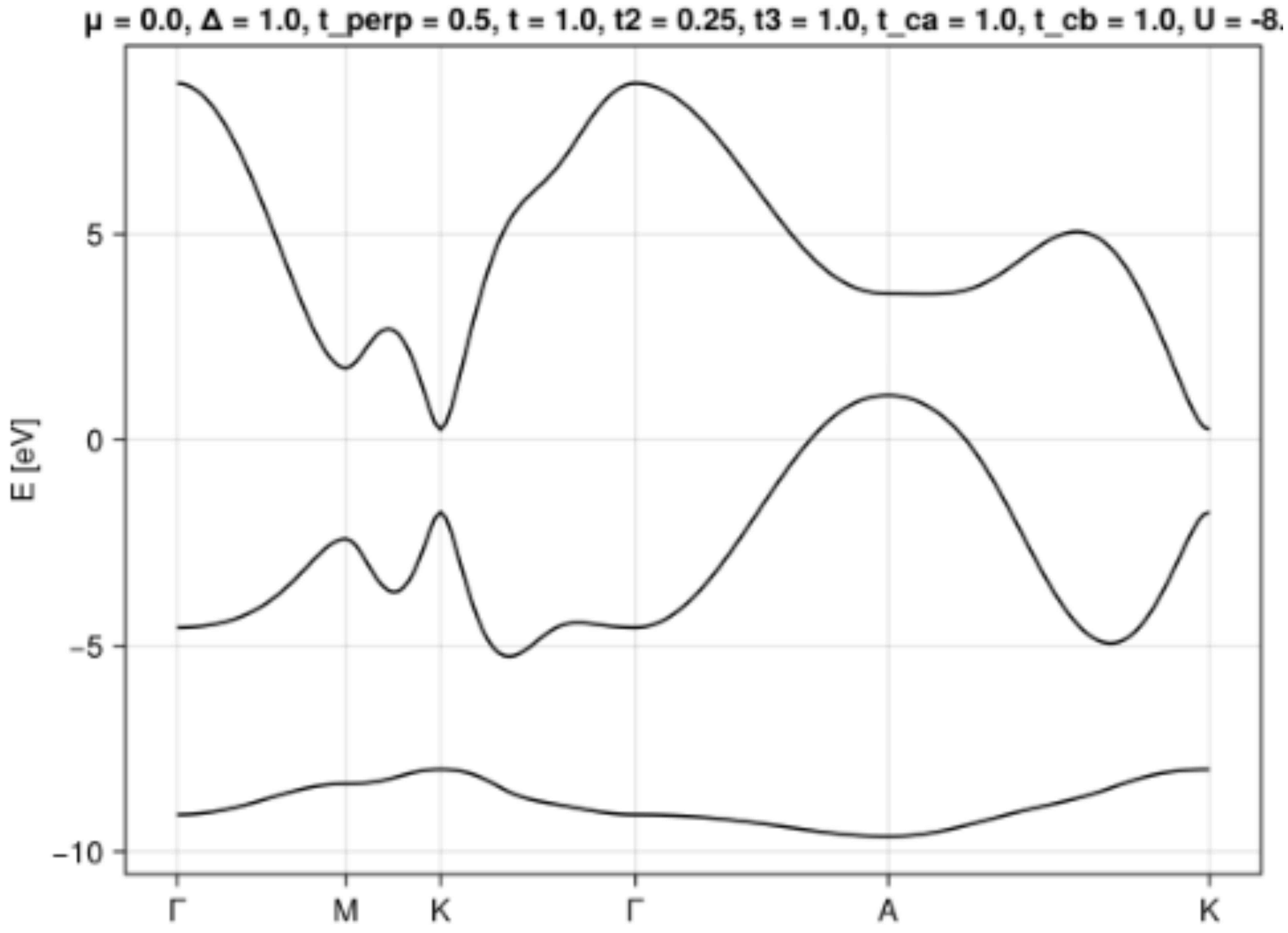
1st $\quad f(k) = \sum e^{i\vec{k}\cdot\vec{\delta}_i}$

2nd $\quad f_2(k) = e^{i\vec{k}\cdot(\vec{\delta}_1-\vec{\delta}_2)} + e^{i\vec{k}\cdot(\vec{\delta}_2-\vec{\delta}_3)} + e^{i\vec{k}\cdot(\vec{\delta}_3-\vec{\delta}_1)} + e^{i\vec{k}\cdot(-\vec{\delta}_1+\vec{\delta}_2)} + e^{i\vec{k}\cdot(-\vec{\delta}_2+\vec{\delta}_3)} + e^{i\vec{k}\cdot(-\vec{\delta}_3+\vec{\delta}_1)}$

3rd (opposite signs) $\quad g(\vec{k}) = e^{i\vec{k}\cdot(2\vec{\delta}_1-\vec{\delta}_2)} + e^{i\vec{k}\cdot(2\vec{\delta}_2-\vec{\delta}_3)} + e^{i\vec{k}\cdot(2\vec{\delta}_3-\vec{\delta}_1)} - e^{i\vec{k}\cdot(2\vec{\delta}_1-\vec{\delta}_3)} - e^{i\vec{k}\cdot(2\vec{\delta}_2-\vec{\delta}_1)} - e^{i\vec{k}\cdot(2\vec{\delta}_3-\vec{\delta}_2)}$

Perpendicular $\quad m(k) = \sum e^{i\vec{k}\cdot(\vec{\delta}_i+\vec{a}_3)} + \sum e^{i\vec{k}\cdot(\vec{\delta}_i-\vec{a}_3)}$

The three band ferroaxial model - Spectrum



Parameters

$\Delta = 1$ eV

$t = t_3 = t_{CA} = t_{CB} = 1$ eV

$t_\perp = 0.5$ eV

t_2 = 0.25 eV

$a_0 = c = 1\text{Å}$

$\mu = 0$ eV

$U = -8$ eV